53.6% of patients achieved a score classified as high quality of life, while 46.4% did not reach this threshold. Thus, more than half of the patients reported an overall positive perception despite the complexity of their condition, yielding results comparable to those reported in other cohorts.

Conclusion: Most patients reported a high perceived quality of life. Nevertheless, long-term follow-up and the prevention or management of late complications should be prioritized, as these factors may play a critical role in sustaining and further improving quality of life.



MO 18
ROBOTIC CHOLECYSTECTOMY LEARNING AND TEACHING CURVES FOR ATTENDINGS

G. Pradhan, P. Krzesaj, J. Bistre-Varon, C. King, A. Nigam, V. Davidyuk and S. DiBrito

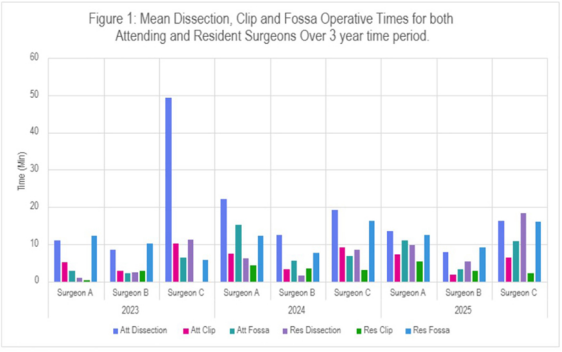
Presenter: Grishma Pradhan MD | Albany Medical Center

Background: Robotic cholecystectomy (RC) is one of the most frequently performed robotic procedures and often serves as an introductory operation for surgical trainees due to relatively standardized anatomy and lower risk profile. Prior studies have demonstrated varying learning curves depending on learners' prior laparoscopic experience, as well as multiport vs single site operative approach. It remains unclear how many cases are needed prior to attendings allowing residents to perform RC independently.

Methods: To identify time needed to attain proficiency in RC, Intuitive data from the Da Vinci system was obtained from 369 RC cases performed by three attending surgeons at a large academic teaching hospital in NY between the years 2016 − 2025. Data collected included: whether a resident participated or not and their respective level, total operating time, time spent on the different phases of the surgery such as dissection, clipping the duct and artery and dissecting the gallbladder off of the fossa. Statistical Testing included both t test and chi-squared test.

Results: Each attending's cases were analyzed independently. Surgeon A completed 133 cases between 2016 and 2025, Surgeon B completed 173 cases between 2022 and 2025, and Surgeon C completed 63 cases between 2023 and 2025. Comparison of operative times for cases performed by the attending alone versus with resident participation demonstrated no significant differences across all three surgeons (Surgeon A: solo mean 42.42 min vs. resident-assisted mean 39.14 min, $p = 0.5587$; Surgeon B: solo mean 30.39 min vs. resident-assisted mean 24.33 min, $p = 0.4381$; Surgeon C: solo mean 70.11 min vs. resident-assisted mean 60.12 min, $p = 0.4632$). Analysis of resident participation over time revealed significant increases for Surgeon A and Surgeon B (Surgeon A: $\chi^2(8, N = 18) = 28.51$, $p < 0.0004$; Surgeon B: $\chi^2(3, N = 8) = 12.61$, $p < 0.005$), indicating a statistically significant upward trend in resident involvement across successive years. Figure below shows an overall trend in increasing resident participation in each phase, without showing statistical significance.

Conclusion: Overall, our analysis demonstrates that resident participation does not result in significantly longer operative times. Among the surgeons with higher case volumes, resident involvement showed significant differences over the study period. We plan to expand this analysis by including additional surgeons to determine whether there is an overall trend of residents receiving more operative time as attendings become more comfortable working with them. Furthermore, we aim to evaluate differences in resident participation across distinct phases of surgery. Although we observed a trend of increasing resident involvement in each phase over successive years, these differences did not reach statistical significance.



MO 19
COMPARING SURVIVAL AND RECURRENCE IN HILAR CHOLANGIOCARCINOMA PATIENTS AFTER ORTHOTOPIC LIVER TRANSPLANTATION OR ANATOMICAL HEPATIC RESECTION

A. Giuseppucci, S. Shoucair, P. Bansal, M. A. Nadeem, J. K. Kim, D. Kwon, F. Aucejo and M. Khalil

Presenter: Antonio Giuseppucci BS | Cleveland Clinic Foundation

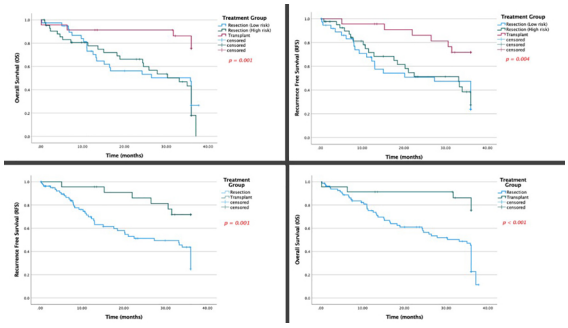
Background: Hilar Cholangiocarcinoma is an aggressive malignancy with an overall 5-year survival rate of about 9% and a reported recurrence rate of 76%. Treatment options include chemoradiation and surgery. Surgical resection is considered standard treatment in patients with resectable disease. Traditionally, orthotopic liver transplantation has been reserved for carefully selected unresectable cholangiocarcinoma patients following neoadjuvant chemoradiation. We aimed to compare survival outcomes in resected and transplanted hilar cholangiocarcinoma patients, hypothesizing a benefit in survival with liver transplant that may suggest an expansion of current transplantation indications.

Methods: We analyzed a retrospective database of cholangiocarcinoma patients treated by anatomical hepatic

resection or orthotopic liver transplantation from 2015-2024. We compared recurrence free survival and overall survival at 3 years using Kaplan-Meier analysis. Resected patients were further stratified into "high risk" (Bismuth classification 4 and/or lympho-vascular invasion) and "low risk" groups, and their recurrence and overall survival was compared to transplanted patients. Data were analyzed using IBM SPSS Statistics (IBM Corp.).

Results: Our database included 105 patients (60% Male) with a median age of 62.9 ± 12.8 years, a median BMI of 25.2 ± 3.1 kg/m2. Median follow-up was 30.1 months (10.7 - 36). 82 (78.1%) were resected, while 23 (21.9%) were transplanted. Between the two groups, no differences in gender or BMI were observed. Patients were significantly older in the resected group (65.2 vs 52.0 years, p = .005). Transplanted patients were also significantly more likely to have a history of primary sclerosing cholangitis (87.0% vs 13.4% p < .001). Kaplan-Meier analysis showed an improved recurrence-free survival (p = 0.001) and overall survival (p < 0.001) within 3 years of surgery for transplant patients (mRFS = 48.5 mo, mOS = 61.1 mo) when compared to resected patients (mRFS = 27.3 mo, mOS = 33.1 mo). Further analysis comparing transplanted patients to "high risk" and "low risk" resection cohorts showed an improved recurrence-free survival (p = 0.001) and overall survival (p < 0.001) within 3 years of surgery for both groups.

Conclusion: Hilar cholangiocarcinoma's dismal recurrence and survival rates inform the medical community's efforts to optimize treatment for this disease. Our results indicate that transplantation may lead to more positive prognoses for this patient population when compared to surgical resection, regardless of the degree of disease, and warrant exploration into broadened indications for transplantation of hilar cholangiocarcinoma. Further work into recurrence and survival risk-score optimization is needed.



MO 20
ROBOTIC RE-OPERATIVE SURGERY FOR RECURRENT GALLBLADDER DISEASE: A CONTEMPORARY SINGLE-INSTITUTION EXPERIENCE

S. Nahi, H. Alkhafaji, M. Porembka, S. Wang, A. Yopp, P. Polanco, J. Mansour, H. Zeh and C. Ethun

Presenter: Skylar Nahi MD | University of Texas Southwestern Medical Center

Background: Incomplete or partial cholecystectomy is increasingly performed in challenging operative settings owing to extent of disease, prior surgical history, or body habitus. Residual gallbladder or cystic duct pathology can lead to recurrent symptoms requiring reoperation. Re-operative gallbladder surgery is technically challenging due to adhesions, distorted anatomy, and increased risk of bile duct injury. The role of a robotic surgical approach in this setting remains poorly defined.

Methods: We conducted a retrospective cohort study of patients undergoing robotic re-operative cholecystectomy for recurrent gallbladder disease at a tertiary academic center (2022−2024). Demographics, comorbidities, operative details, imaging findings, perioperative outcomes, and long-term follow-up were reviewed. Outcomes were compared to institutional experience with primary cholecystectomy during the same period.

Results: Twenty-one patients underwent robotic re-operative gallbladder surgery. Median age was 41 years, 91% were female, 91% were Hispanic, and 71% were obese. Most index operations were laparoscopic (67%), and 40% were documented as partial/subtotal resections. At re-presentation, diagnostic imaging identified a "remnant gallbladder" or dilated cystic duct in 100% of patients, cystic duct stones in 67%, and choledocholithiasis in 24%. Median operative time was 166 minutes; no intraoperative complications occurred. Median postoperative hospital stay was 0 nights (range 0−2). Thirty-day outcomes included one readmission for major complication: formation of an intraabdominal abscess. This individual patient was managed conservatively without reoperation. At follow up, 100% had resolutions of symptoms. No 90-day major morbidity occurred. Kaplan−Meier failure function analysis demonstrated a steady incidence of reoperation after index cholecystectomy. Compared to >2,000 primary cholecystectomies at the same institution, this robotic re-operative cohort demonstrates acceptable safety and short hospitalizations despite technical complexity.

Conclusion: Robotic re-operative cholecystectomy for recurrent gallbladder disease is safe, feasible, and associated with minimal postoperative morbidity and short hospital stay. In the era of increasing subtotal cholecystectomy, robotic platforms may offer distinct advantages in managing this challenging re-operative population. Further comparative studies are warranted to define optimal operative strategies.

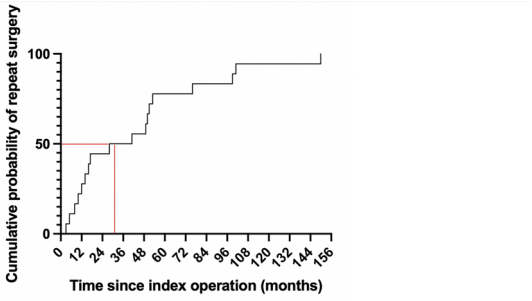


Figure 1. Kaplan-Meier failure function estimating likelihood of repeat operation as a function of time from index operation. The y-axis represents the cumulative probability of having undergone surgery, and the x-axis represents time in months from index operation. Median: 34.5 months. *Patients who did not have a documented date of index surgery (n=3) were censored.*